Rongwu Xu

Personal

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Research

My primary interests lie in the field of **natural language processing (NLP)**. My current exploration focuses on the following aspects of **large language models (LLMs)**:

- *Data:* Working on data-driven learning, synthetic data, data generation and data assessment. Data is everywhere in LLMs' pipelines.
- *Evaluation:* Constructing reliable frameworks and datasets to assess LLMs' language generation, contextual understanding, factuality, robustness, among others.
- *Safety:* Identifying risks in LLMs such as misinformation and ethical issues, aiming to build NLP systems that reflect human values.

I have supervised a total of **10** undergraduate/master's students, many of whom have co-published articles under my mentorship.

Education

Jun. 2025	MRes in Computer Science **Tsinghua University** Beijing, China Advisor: Prof. Wei Xu
Aug. 2022	Graduate Student at IIIS (Headed by Andrew C. Yao, the Turing award laureate'2000)
Jun. 2022	BEng in Computer Science **Tsinghua University** Beijing, China
Sept. 2018	Bachelor Student at Department of Computer Science and Technology (DCST)
Jun. 2018	**Beijing No.4 High School** Beijing, China
Sept. 2015	High School Student at the Class of Olympiad (Chemistry)

Publications and Manuscripts

TL;DR To date, I have authored **16** research papers, including **10** first/co-first author papers. A total of **8** papers have been published in international conferences and journals.

1. DebateQA: Evaluating Question Answering on Debatable Knowledge
   Rongwu Xu*, Xuan Qi*, Zehan Qi, Wei Xu, Zhijiang Guo
   Preprint

2. Course-Correction: Safety Alignment Using Synthetic Preferences
   Rongwu Xu*, Yishuo Cai*, Zhenhong Zhou, Renjie Gu, Haiqin Wang, Yan Liu, Tianwei Zhang, Wei Xu, Han Qiu
   arXiv Preprints

3. LONG^{2}RAG: Evaluating Long-Context & Long-Form Retrieval-Augmented Generation with Key Point Recall
   Zehan Qi*, **Rongwu Xu***, Zhijiang Guo, Cunxiang Wang, Hao Zhang, Wei Xu
   Preprint

4. MR-BEN: A Comprehensive Meta-Reasoning Benchmark for Large Language Models
   Zhongshen Zeng, Yinhong Liu, Yingjia Wan, Jingyao Li, Pengguang Chen, Jianbo Dai, Yuxuan Yao, **Rongwu Xu**, Zehan Qi, Wanru Zhao, Linling Shen, Jianqiao Lu, Haochen Tan, Yukang Chen, Hao Zhang, Zhan Shi, Bailin Wang, Zhijiang Guo, Jiaya Jia
   arXiv Preprints

5. How Alignment and Jailbreak Work: Explain LLM Safety through Intermediate Hidden States
   Zhenhong Zhou, Haiyang Yu, Xinghua Zhang, **Rongwu Xu**, Fei Huang, Yongbin Li
   arXiv Preprint

6. Sing it, Narrate it: Quality Musical Lyrics Translation
   Jinhan Li, Zhuorui Ye, **Rongwu Xu**
   Preprint

7. Walking in Others' Shoes: How Perspective-Taking Guides LLMs in Reducing Toxicity and Bias
   Rongwu Xu, Zi'an Zhou, Tianwei Zhang, Zehan Qi, Su Yao, Ke Xu, Wei Xu, Han Qiu
   Preprint

8. Knowledge Conflicts for LLMs: A Survey
   Rongwu Xu*, Zehan Qi*, Cunxiang Wang, Hongru Wang, Yue Zhang, Wei Xu
   arXiv Preprint

Below are published papers

9. Preemptive Answer "Attacks" on Chain-of-Thought Reasoning
   Rongwu Xu*, Zehan Qi*, Wei Xu
   In Findings of *the 62nd Annual Meeting of the Association for Computational Linguistics* (**ACL**-Findings), 2024

10. The Earth is Flat because...: Investigating LLMs' Belief towards Misinformation via Persuasive Conversation
    Rongwu Xu, Brian S. Lin, Shujian Yang, Tianqi Zhang, Weiyan Shi, Tianwei Zhang, Zhixuan Fang, Wei Xu, Han Qiu
    In Proceedings of *the 62nd Annual Meeting of the Association for Computational Linguistics* (**ACL**-Main, **Oral**), 2024

11. Exploring Chinese Humor Generation: A Study on Two-part Allegorical Sayings
    Rongwu Xu
    In Proceedings of *International Joint Conference on Neural Networks* (**IJCNN**), 2024

12. Tempo: Confidentiality Preservation in Cloud-based Neural Network Training
    Rongwu Xu and Zhixuan Fang
    In Proceedings of *International Joint Conference on Neural Networks* (**IJCNN**), 2024

13. LSync: A Universal Timeline-synchronizing Solution for Live Streaming
    Yifan Xu*, Fan Dang*, **Rongwu Xu**, Xinlei Chen, Yunhao Liu
    In *IEEE/ACM Transactions on Networking* (**ToN**), 2024

14. MISO: Legacy-compatible Privacy-preserving Single Sign-on using Trusted Execution Environments
    Rongwu Xu, Sen Yang, Fan Zhang, Zhixuan Fang
    In Proceedings of *IEEE European Symposium on Security and Privacy* (**EuroS&P**), 2023

15. LSync: A Universal Event-synchronizing Solution for Live Streaming
    Yifan Xu, Fan Dang, **Rongwu Xu**, Xinlei Chen, Yunhao Liu
    In Proceedings of *IEEE Conference on Computer Communications* (**INFOCOM**), 2022
16. LifeRec: A Mobile App for Lifelog Recording and Ubiquitous Recommendation
    Jiayu Li, Hantian Zhang*, Zhiyu He*, **Rongwu Xu***, Pingfei Wu*, Min Zhang, Yiqun Liu, Shaoping Ma
    In Proceedings of *ACM SIGIR Conference on Human Information Interaction and Retrieval* (**CHIIR**), 2022

(* equal contribution)

Honors and Awards

2023	Overall Excellence Scholarship@Tsinghua University
2020	Technological Innovation Scholarship@Tsinghua University
2020	1st in "Youth in Action" Social Practice@Tsinghua University
2019	Tsinghua-Panasonic Scholarship@Tsinghua University
2018	Outstanding Volunteers in Beijing
2017	2nd Prize (Preliminary) in Chinese Chemistry Olympiad (CChO)
2017	1st Prize in Chinese Chemistry Olympiad (Beijing Regional Qualifiers)

Talks and Presentations

Aug. 2024	Investigating LLMs' beliefs and behaviors under misinformation	Oral report@ACL conference
Jul. 2024	Knowledge conflicts for (RAG) LLMs	Online talk@NICE
Apr. 2024	Investigating LLMs' beliefs and behaviors under misinformation	Propaganda film@IIIS, Tsinghua
May. 2023	Privacy-preserving authentication	Oral report@EuroS&P conference

Experiences

Mentoring

It is my pleasure to collaborate with the following brilliant students:

Apr. 2024 -	Yishuo Cai	Undergrad, SE@Central South Univ.
Mar. 2024 -	Yishu Yin	Undergrad, CS@Tsinghua Univ.
Mar. 2024 -	Priscilla Chen	Undergrad, EECS@UC Berkeley
Mar. 2024 - Apr. 2024	Xinghan Li	Undergrad, IIIS@Tsinghua Univ.
Mar. 2024 -	Xuan Qi	Undergrad, IIIS@Tsinghua Univ.
Dec. 2023 - May. 2025	Zi'an Zhou	Undergrad, Zhili College@Tsinghua Univ.
Sept. 2023 - Feb. 2024	Shujian Yang	Master, SPEIT@Shanghai Jiao Tong Univ.
Jun. 2023 - Apr. 2024	Tianqi Zhang	Undergrad, CS@Tsinghua Univ.
Jun. 2023 - Apr. 2024	Brian S. Lin	Undergrad, CS@Tsinghua Univ.
Oct. 2021 - Jul. 2022	Xingyu Dang	Undergrad, IIIS@Tsinghua Univ.

Teaching

Spring 2024	**Lead Teaching Assistant & Organizer**, Tsinghua University *Introduction of Large Language Model Applications* Directed Co-designed labs by mobilizing 10 graduate students, organized the curriculum and assisted labs in class.
Fall 2023	**Teaching Assistant**, Tsinghua University *Operating System and Distributed System* Held discussion and office hours, graded exams, assignments and projects.
Spring 2022	**Teaching Assistant**, Tsinghua University *Distributed System and Blockchain* Held discussion and office hours, graded exams, assignments and projects.

Exchanging

Apr. 2021 - Oct. 2022	**Research Assistant** (Remote), Duke University *Granted summer overseas internship (undergrad) by Tsinghua* Conducted research in privacy-preserving authentication. The research findings were published in the EuroS&P conference. Host: Prof. Fan Zhang

Internship

May. 2024 -	**TongYi Vision Intelligence Lab, Alibaba Inc.**, Beijing, China Conducted research in visual language tasks and world model. Mentor: Yu Liu
Dec. 2022 - Jan. 2023	**Shanghai Qi Zhi Institute**, Shanghai, China Conducted research in decentralized finance (DeFi). Worked on MEV arbitrage forecasting algorithms using graph neural networks (GNNs). Mentor: Prof. Zhixuan Fang

Skills and Expertise

- **Research:** Experienced in deep learning/data analysis, also ability with computer systems/applied cryptography/software engineering.
- **Programming Language:** Proficient in C++/Go/Python/Java/JavaScript/LaTeX, also ability with Verilog/System Verilog/ASM/Rust/P4/HTML/Matlab.
- **Technological:** Proficient in Pytorch/NumPy/Matplotlib/Git/Linux OS/Markdown/, also ability with Docker/Wireshark/Microsoft Office.
- **Other Expertise:** Good communication skills, love collaborating with people, experienced in mentoring students, works well in a team.
- **Standard Language Test (TOEFL):** 106.

Services and Memberships

Academic

Mar. 2024 - Current	**Member**, International Neural Network Society (INNS)
Mar. 2024 - Current	**Member**, Institute of Electrical and Electronics Engineers (IEEE)
Mar. 2024 - Current	**Member**, ACL SIGSEC, Association for Computational Linguistics

Societal

Jun. 2024 - Current	**President**, IIIS Graduate Students Union, Tsinghua University In charge of organization and publicity.
May. 2024 - Current	**Freshman Counsellor**, IIIS, Tsinghua University *Class 2024 (graduate)* Responsible for the communication of new students and the collective formation of class.
Apr. 2024 - Jul. 2024	**Captain**, summer social practice (graduate, Shenzhen—Hong Kong line), IIIS, Tsinghua University Responsible for organization, mobilization, liaison (government) and advocacy.
Sept. 2023 - Jun. 2024	**Member**, IIIS Graduate Students Union, Tsinghua University Participated in the organization of the first student festival and other social activities. Awarded *excellent individual* of IIIS Graduate Students Union (2023—2024)

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